

Echofish

An envelope and step filter with a tape, bucket-brigade or digital echo below it.



Overview

Echofish is two effects on one panel. The top is an automatic filter in the spirit of the classic 70s envelope-filter pedals and the modern sample-and-hold filter machines: a resonant two-pole filter that is swept by your playing (the classic auto-wah quack) or stepped in time with the project.

The bottom strip is an echo with three characters - warm, saturating TAPE, darker BUCKET (bucket-brigade) and clean DIGITAL - with sync, ping-pong, freeze, ducking and a vintage EQ on the repeats.

ORDER sets which comes first: FILTER > ECHO, so the echoes repeat the filtered sound, or ECHO > FILTER, so the filter moves over the echoes too.

Reach for it for

- Funky envelope-filter guitar, clav and bass
- Rhythmic sample-and-hold filter steps locked to the song

- Dub-style echoes that wobble and saturate (TAPE, high FEEDBACK)
- Filtered synth blips and gurgles with a ping-pong echo
- Vocal or lead delays that duck out of the way while the part plays (DUCK)

Controls

The header

HOLD	Freezes the filter where it is. The cutoff stops moving until you switch HOLD off.
ON	Switches the filter in or out. With it out, the dry signal passes through the filter section; the echo keeps working.
PRESETS	Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use (click it for the list too). Choosing one sets the filter controls; the echo and the input / output strip are left as they are. The last entry resets the filter and the IN / OUT strip to their defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Display

Cutoff trace	The filter's cutoff over the last few moments, on a 25 Hz - 18 kHz scale, with its current value at the top right. In S&H modes a faint line marks each new step.
ENV	Lights when the envelope follower picks up your playing.

Sweep

Up / down arrows	The sweep direction. Click the up arrow (blue block, left) for UP: the filter opens upwards from FREQ. Click the down
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	arrow (red block, right) for DOWN: it sweeps down from FREQ. The block in use is lit.
DEPTH	How far the sweep goes, 0 - 6 octaves.
FREQ	Where the sweep starts, 60 Hz - 3 kHz.
FILTER	The filter type: LP (low-pass), BP (band-pass, the classic wah), HP (high-pass) or N (notch).
RANGE LO / HI	HI moves the whole sweep up an octave.
PATTERN FREE / LOOP	For the S&H modes. FREE: a new random step every time. LOOP: a one-bar random pattern that repeats, so it grooves with the song.
ENV / S&H / BOTH	What moves the filter. ENV: the envelope follower - the harder you play, the further it sweeps. S&H: random steps in time with the project. BOTH: the steps, scaled by how hard you play.

The knob row

PEAK	The filter's resonance, 0.5 - 15. Higher is more vocal and whistling.
SENS	How strongly the envelope follows your playing. Raise it for quiet sources, lower it if the filter sits fully open.
RESPONSE	How fast the envelope follows: from fast and snappy (about 1 ms attack, 30 ms release) to slow and vocal (about 20 ms attack, 800 ms release).
RATE	The step length for S&H: 1/2, 1/4, 1/8, 1/16 or 1/32 of a note.
MULT	Multiplies the step rate by 1, 2 or 4: x2 gives twice as many steps per RATE.
GLIDE	Smooths the jump between steps, 0 - 100%. Higher settings slide from one step to the next.
DRIVE	Pushes the filter for a fatter, grittier sound, 0 - 100%.

MIX	Dry / filtered balance, 0 - 100%.
OUTPUT	The filter's output level, -12 to +12 dB.

The steps follow the project's position while it plays, and keep running at the project tempo when it is stopped (120 BPM if there is no tempo).

Echo

SYNC	On (default): TIME is a note value at the project tempo. Off: TIME is in milliseconds.
PING-PONG	The input enters on the left and the echoes bounce from side to side.
FREEZE	Holds the echoes exactly as they are and lets no new sound in: an endless loop. Switch it off to let it fade.
VINTAGE	Click to step through off, VINTAGE -3, -6 and -9. On, the echoes you hear get the classic studio send EQ: a high-pass at 140 Hz, a low-pass at 5 kHz and a 3, 6 or 9 dB cut at 500 Hz. The tone inside the feedback loop is still set by LO CUT and HI CUT.
ORDER	FILTER > ECHO or ECHO > FILTER: click to swap.
TAPE / BUCKET / DIGITAL	The echo's character. TAPE: soft saturation and a gently rolled-off top; changing TIME bends the pitch slowly, like a tape machine. BUCKET: darker, with a firmer clip and a slight chorus on the tails. DIGITAL: clean, with only a safety limit on the feedback.
TIME	The echo time: 20 - 2000 ms, or with SYNC from 4 bars down to 1/32, including dotted (D) and triplet (T) values. 1/8 D by default.
FEEDBACK	How much each echo feeds the next, 0 - 110%. Above 100% the echoes build up and run away (TAPE and BUCKET saturate to keep it in check).
DRIVE	How hard the feedback loop saturates in TAPE and BUCKET, 0 - 100%. DIGITAL stays clean.

LO CUT / HI CUT	Filters inside the feedback loop, so each repeat gets thinner and darker: LO CUT 20 - 1000 Hz, HI CUT 1 - 20 kHz.
DUCK	Lowers the echoes while the input is playing, 0 - 100%, so they fill the gaps rather than the part.
DRIFT	The tape's wow and flutter, 0 - 100% (50% is the standard amount, 100% twice that, 0% none). It wobbles the echo time, and the wander speeds up a little as you turn it up.
WIDTH	The stereo width of the echoes, 0 - 100%. Without PING-PONG, the right echo is also very slightly longer at full width.
MIX	Dry / echo balance, 0 - 100%. The dry signal stays at full level up to 50%, then fades; at 100% you hear only the echoes.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob to switch it; its LED sits at the knob's upper right. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The LED glows brighter while it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). CLIP is on by default and flashes red when it clips.

In REAPER's parameter list only

EQ: Wow / EQ: Flutter	The standard amounts of wow and flutter that DRIFT scales (20% and 15% by default).
EQ: Drift speed	How fast the wow wanders, 0.05 - 2 Hz.
EQ: Output	The echo section's output level, -24 to +12 dB.
Bias / Sat low band / Sat high band / Ceiling	The rest of the output stage: the saturation's even-harmonic bias, where its low and high bands start (40 - 400 Hz and 2 - 16 kHz) and the clip ceiling or limiter threshold.

Presets

Init	The default filter settings: an upward band-pass envelope wah from 250 Hz. A new instance opens on it.
Funk Up	The classic up-sweeping envelope wah: band-pass from 250 Hz, a little drive.
Bass Quack (down)	Low-pass sweeping down from 900 Hz, for bass.
Ship Steps	A looping one-bar pattern of 1/16 steps through a resonant band-pass.
Random Gurgle	Random, never-repeating steps (1/8 × 2) with a lot of glide through a resonant low-pass.
Envelope Steps	A looping 1/16 pattern, scaled by how hard you play.
Notch Sweep	An envelope-driven notch with a slower response.
Slow Talker	A very resonant band-pass with a slow, vocal response.
Clav Wah	Fast, bright band-pass wah in the HI range with extra drive.
Synth Blips	Very resonant 1/32 steps in the HI range, no glide.
Lowpass Bloom	A gentle low-pass opening five octaves up from 120 Hz with a slow response.

Reset all to defaults

The filter and the IN / OUT strip back to their starting values; the echo is left as it is.

Tips

- If the envelope filter barely moves, raise SENS; if it sits wide open, lower it or turn TRIM down.
- S&H with PATTERN on LOOP gives a riff that repeats every bar; a little GLIDE makes the steps sound less mechanical.
- For dub throws, use TAPE with FEEDBACK around 85 - 100% and automate MIX; FREEZE holds the last phrase while you play over it.
- ECHO > FILTER lets each echo get swept as well, which sounds great with S&H on sustained parts.
- DUCK at 30 - 50% keeps a busy echo out of the way of a vocal or lead.

File: gfx-echofish.jsfx